Check for updates

*Advanced Quantum Technologies*

ADVANCED QUANTUM TECHNOLOGIES
www.advquantumtech.com

RESEARCH ARTICLE OPEN ACCESS

# Hybrid Quantum–Classical Graph Transformers for Efficient Sentiment Analysis

Shamminuj Aktar[1] | Andreas Bärtschi[1] | Abdel-Hameed A. Badawy[2] | Stephan Eidenbenz[1]

[1]Computing and Artificial Intelligence Division (CAI-3), Los Alamos National Laboratory, Los Alamos, USA | [2]Knight Foundation School of Computer & Information Sciences, Florida International University, Miami, USA

**Correspondence:** Shamminuj Aktar (saktar@lanl.gov)

**Received:** 24 November 2025 | **Revised:** 2 March 2026 | **Accepted:** 2 March 2026

**Keywords:** graph transformer | self-attention mechanism | text classification | quantum machine learning

## ABSTRACT

Quantum Machine Learning (QML) offers a promising paradigm that leverages quantum computing principles to develop efficient and expressive models for learning from complex and structured data. Recent advances in natural language processing (NLP) and artificial intelligence (AI) have demonstrated capabilities in understanding, generating, and reasoning over linguistic and multimodal information. In this work, we present the *Quantum Graph Transformer (QGT)*, a hybrid quantum–classical architecture that extends graph transformer capabilities through quantum self-attention. The QGT models variable-length sentences as token graphs, where both the embedding encoding and the self-attention mechanisms are implemented using parameterized quantum circuits (PQCs), enabling efficient contextual learning with significantly fewer trainable parameters. We train QGT using both fully connected and $k$-nearest-neighbor graph structures and evaluate it on five benchmark sentiment-classification datasets. Experimental results show that QGT consistently achieves higher or comparable accuracy to existing quantum NLP models and outperforms a Classical Graph Transformer (CGT) baseline with identical architecture, achieving 29.4× fewer parameters while requiring 3–5× fewer samples to reach comparable performance. These findings highlight the potential of graph-based quantum models as scalable and data-efficient architectures for natural language understanding.

## 1 | Introduction

Artificial Intelligence (AI) has become an integral component of modern technological progress, influencing diverse domains such as healthcare, finance, scientific discovery, and communication [1]. Among its subfields, Natural Language Processing (NLP) plays a foundational role by enabling machines to understand, interpret, and generate human language [2]. This capability powers a wide range of applications, including sentiment analysis, machine translation, conversational agents, and content moderation [3, 4]. Recent advancements in large language models, such as GPT-4, have substantially improved the ability of machines to produce coherent and context-aware text [5]. Despite these achievements, current NLP models face several challenges. Training and inference for state-of-the-art systems demand vast amounts of labeled data, high-performance computing (HPC) infrastructure, and the optimization of billions of parameters [6].

Quantum Machine Learning (QML) offers a promising framework to address some of the inefficiencies inherent in classical machine learning (ML) [7, 8]. QML employs a hybrid quantum–classical paradigm in which data is encoded into a high-dimensional Hilbert space and processed through Parameterized Quantum Circuits (PQCs). These PQCs act as trainable layers analogous to those in classical neural networks, while leveraging quantum properties such as superposition and entanglement to explore complex functions more efficiently [9, 10]. In NLP, where modeling contextual dependencies between tokens is essential,

This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2026 The Author(s). Advanced *Quantum Technologies* published by Wiley-VCH GmbH

QML provides a pathway toward developing more efficient and expressive language models. Quantum-enhanced NLP (QNLP) approaches have the potential to offer quantum advantages in sample efficiency, model accuracy, convergence speed, and generalization capability.

Recent research in QNLP has explored multiple strategies for integrating quantum principles into language modeling. Early models include quantum bag-of-words representations, quantum support vector machines (QSVMs), and variational quantum classifiers based on amplitude encoding [11, 12]. More recent developments have introduced hybrid architectures such as quantum-enhanced LSTMs [13], self-attention-based models [14–16], and ensemble-based frameworks like LexiQL [17]. These studies have demonstrated promising results, suggesting that QNLP can serve as a viable and scalable alternative to classical NLP approaches. In parallel, graph-based models in classical machine learning have shown strong capability in representing complex structured data, including knowledge graphs, citation networks, and syntactic dependency trees in NLP [18, 19]. Graph Neural Networks (GNNs) model data as graphs, where nodes represent entities (such as tokens in a sentence) and edges capture syntactic or semantic relationships [20]. This graph-based representation enables models to capture both hierarchical and contextual relationships effectively.

Building upon these insights, we propose the Quantum Graph Transformer (QGT), a novel hybrid quantum–classical model for sentiment classification. The QGT represents each sentence as a connected graph of tokens and introduces a quantum transformer convolution (`QTransformerConv`) layer that propagates information across the graph through a quantum self-attention mechanism. The proposed QGT embeds tokens within a graph topology, allowing it to exploit both local (connecting each node to its $k$ nearest neighbors) and global (fully connected) relational dependencies between words. Within this framework, each token is first encoded into a quantum state, and PQCs are applied to derive the query and key representations used to compute attention scores between connected nodes. These scores guide the message-passing process, enabling the model to capture contextual dependencies among tokens in a quantum-enhanced feature space. We evaluate the QGT model on five benchmark sentiment-classification datasets and demonstrate its effectiveness in learning contextual representations of text.

The main contributions of this work are summarized as follows:

1. We introduce the QGT, a hybrid quantum–classical model that integrates quantum self-attention into a graph-based message-passing framework for structured language modeling. We validate our approach using both BERT's contextual representations and GloVe embeddings to demonstrate that efficient quantum attention learning generalizes across embedding types.
2. We design a parameter-efficient quantum transformer convolution layer (`QTransformerConv`) that computes attention entirely through PQC-based similarity measures, achieving 29.4× fewer parameters than classical attention while maintaining comparable accuracy.
3. The proposed QGT generalizes self-attention to structured token graphs, achieving both topological expressivity and quantum feature efficiency. QGT supports both fully connected and sparse graph attention patterns, reducing computational complexity from $\mathcal{O}(N^2)$ to $\mathcal{O}(Nk)$, where $N$ denotes the number of tokens (graph nodes) and $k$ is the number of nearest neighbors (average node degree). This formulation also eliminates the need for input padding across variable-length sentences.
4. We evaluate the QGT model on five commonly used sentiment-classification benchmarks, including three real-world datasets (Yelp, Imdb, and Amazon) [21] and two synthetic datasets (MC and RP) [11]. QGT outperforms or matches previous QNLP work due to strong contextual embeddings, optimized training configuration, and effective quantum attention learning. Comprehensive ablation studies demonstrate that each quantum component contributes meaningfully to model performance.
5. We provide fair comparison against a Classical Graph Transformer (CGT) baseline with identical architecture and training configuration, confirming that QGT's improvements are attributable to the quantum attention mechanism. Compared with CGT, QGT achieves an average improvement of 2.6% on real-world datasets while requiring 3–5× fewer training samples to reach comparable performance, demonstrating superior sample efficiency.

## 2 | Background and Motivation

### 2.1 | Natural Language Processing Tasks

Natural Language Processing (NLP) is a specialized domain within AI that aims to bridge the gap between human communication and machine understanding by enabling computational systems to process and generate natural language. Core NLP tasks include text classification, named entity recognition, question answering, and machine translation. These tasks are unified by the need to model complex syntactic and semantic relationships within and between sequences of text. In the sentiment classification task, the objective is to predict the underlying emotion or opinion expressed in a given sentence or document, typically as a categorical label such as positive, negative, or neutral. Sentiment classification is a widely studied problem due to its practical relevance in areas such as customer feedback analysis, user experience assessment, and automated review aggregation.

### 2.2 | Classical NLP Models

In the early stages of natural language processing, models primarily relied on simple representations such as Bag-of-Words and $N$-gram features, which treated text as unordered collections of word counts without accounting for syntax or semantics [22, 23]. Subsequently, more linguistically informed approaches emerged, utilizing grammar-aware structures such as dependency trees and part-of-speech tags to capture syntactic relationships between words in a sentence. Although these structured models offered improved linguistic insight, they



25119044, 2026, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/qute.202501004 by Universita Di Salerno, Wiley Online Library on [02/07/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

often struggled to generalize effectively across diverse language contexts. A major advancement came with the introduction of word embeddings such as Word2Vec [24] and GloVe [25], which represent words as continuous vectors in a dense embedding space. These embeddings capture semantic similarity based on word co-occurrence patterns, enabling models to generalize better across related linguistic constructs. Models such as recurrent neural networks (RNNs) [26], long short-term memory networks (LSTMs) [27], and gated recurrent units (GRUs) [28] leveraged these embeddings to process sequential data and learn contextual representations of text. However, these models struggled to capture long-range dependencies due to issues like vanishing gradients and limited memory capacity.

Another breakthrough in NLP came with the introduction of the attention mechanism, which enables models to dynamically focus on relevant parts of the input sequence. This concept was central to the development of the Transformer architecture [29], which entirely replaces recurrence with self-attention layers. Models such as BERT [30], GPT [31], and RoBERTa [32] achieved state-of-the-art results across a wide range of NLP tasks. These models typically require large amounts of labeled data, substantial computational resources, and still face challenges in generalizing to unseen linguistic patterns.

### 2.3 | Existing Quantum NLP Models

QML has been extensively studied in recent years due to its potential to surpass the capabilities of classical machine learning and demonstrate quantum advantage. QML typically employs a hybrid quantum–classical approach, where PQCs are used as trainable quantum layers. A PQC is represented as a unitary transformation $U(\boldsymbol{\lambda})$ acting on an input quantum state $|\psi_{\text{in}}\rangle$, where $\boldsymbol{\lambda} = \{\lambda_1, \lambda_2, \ldots, \lambda_p\}$ are tunable real-valued parameters. The resulting output state is given by $|\psi_{\text{out}}\rangle = U(\boldsymbol{\lambda})|\psi_{\text{in}}\rangle$. In the QML framework, classical input data is first encoded into a quantum state, and the PQC is subsequently trained to learn the optimal parameters that minimize the loss function for the target learning task.

There has been growing interest in QNLP models as an alternative to classical NLP models [33]. Hybrid quantum-enhanced NLP models aim to encode natural language into quantum states and optimize parameters to learn semantic relationships between words. Since quantum states reside in high-dimensional Hilbert space, they offer superior expressive power, even with relatively few parameters. The categorical compositional distributional (DisCoCat) model combines distributional semantics with grammatical structure and maps naturally onto quantum tensor product spaces [34]. Building on this framework, Ganguly et al. demonstrated the application of QNLP for sentiment analysis in noisy quantum devices [35]. More recently, Del Castillo et al. conducted a systematic comparative study of various ansätze to analyze how different circuit designs and hyperparameters affect QNLP model performance in text classification tasks [36]. However, the challenge of accurately extracting syntactic structure from sentence inputs limits the adaptability of the DisCoCat model, especially when handling the variability and ambiguity of natural language. Other approaches include quantum-enhanced bag-of-words (BoW) and N-gram models, which encode word occurrences or co-occurrence patterns into quantum states [12]. These models have shown promising results on synthetic datasets, but their practical implementation and performance are often constrained by the qubit requirements, limiting their scalability and real-world applicability. In addition, Quantum LSTM [13] and Quantum SVM [12] models have been proposed to address sentiment classification tasks similar to classical NLP models. Recent efforts have focused on developing quantum-enhanced self-attention mechanisms. These models demonstrated better learning on real-world benchmark datasets [14–16, 37]. Additionally, a recent work by Silver et al. introduced an ensemble-based technique that employs an incremental data injection approach to improve generalization in quantum NLP models for sentiment classification tasks [17].

### 2.4 | Classical Graph Transformers

Classical graph transformers are a class of graph neural network models that incorporate the self-attention mechanism from the original transformer architecture into graph-structured data [29, 38]. Specifically, the graph transformer replaces fixed graph convolution with learned, attention-based message passing. This allows each node to attend to a subset or all other nodes based on learned attention scores. As a result, graph transformers can effectively capture both local and long-range dependencies in the underlying data. Given an undirected, unweighted graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with node features $\{\mathbf{x}_i \in \mathbb{R}^d\}_{i\in\mathcal{V}}$, each node undergoes a feature transformation through three learned linear projections to obtain query, key, and value representations:

$$\mathbf{Q}_i = \mathbf{W}_Q \mathbf{x}_i, \quad \mathbf{K}_j = \mathbf{W}_K \mathbf{x}_j, \quad \mathbf{V}_j = \mathbf{W}_V \mathbf{x}_j. \tag{1}$$

where $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{d\times d}$ are shared trainable parameters. The attention score $e_{i,j}$ between target node $i$ and its neighbor $j$ is computed using a scaled dot product between their query and key vectors:

$$e_{i,j} = \frac{\mathbf{Q}_i^\top \mathbf{K}_j}{\sqrt{d}}. \tag{2}$$

To ensure these scores are comparable across neighbors, they are normalized using the softmax function:

$$\alpha_{i,j} = \frac{\exp(e_{i,j})}{\sum_{m\in\mathcal{N}(i)} \exp(e_{i,m})}, \tag{3}$$

where $\mathcal{N}(i)$ denotes the set of neighbors of node $i$. The resulting attention coefficients $\alpha_{i,j}$ reflect the relative importance or relevance of each neighbor $j$ when updating node $i$'s feature vector. The final output feature for node $i$ is computed as a weighted sum of the value vectors of its neighbors:

$$\mathbf{x}_i' = \mathbf{x}_i + \sum_{j\in\mathcal{N}(i)} \alpha_{i,j} \mathbf{V}_j. \tag{4}$$

This formulation enables each node to selectively focus on different parts of its neighborhood, allowing the model to dynamically encode structural information based on learned relevancy scores. Multiple layers of transformer-based message passing can be



25119044, 2026, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/qute.202501004 by Universita Di Salerno, Wiley Online Library on [02/07/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

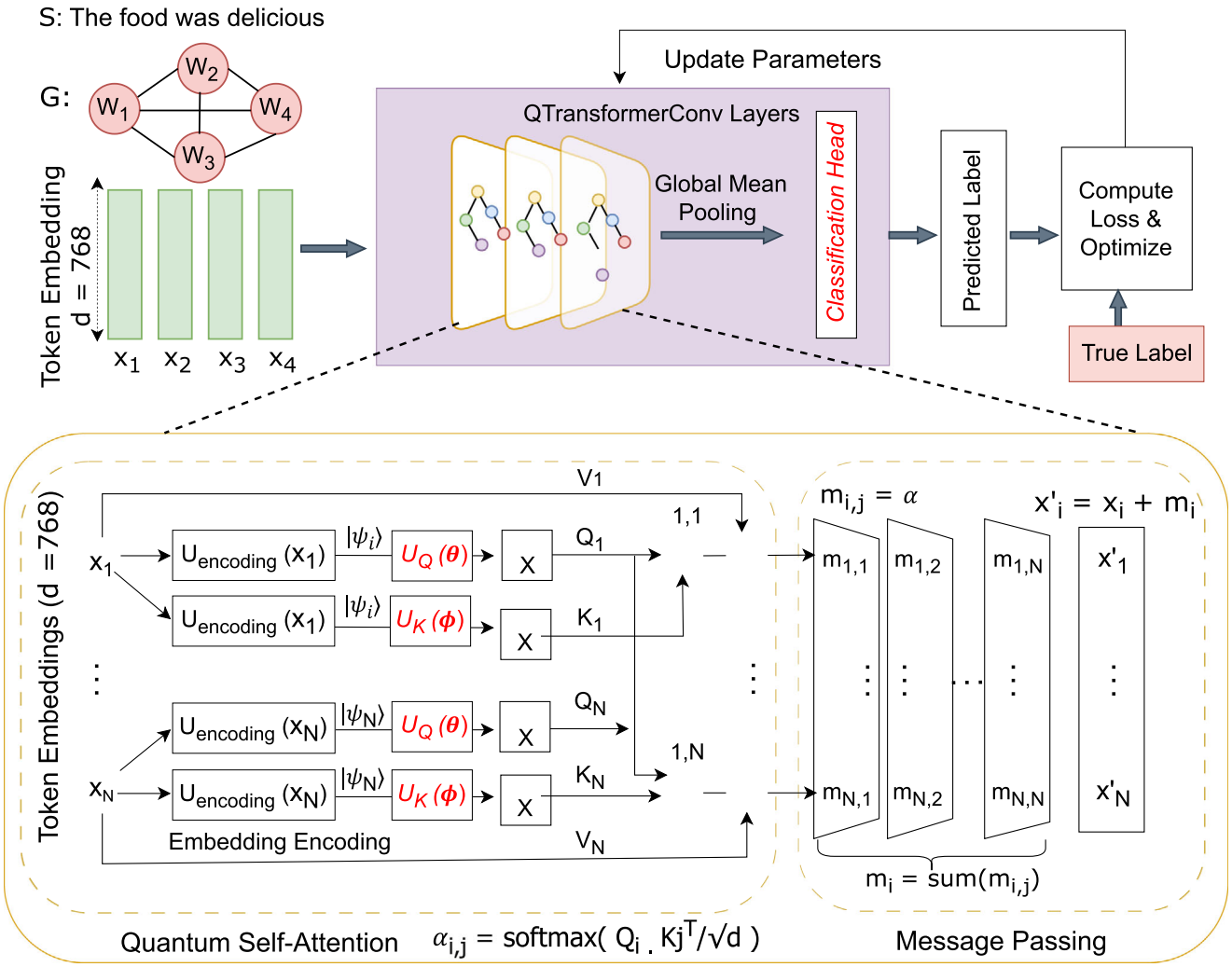



**FIGURE 1** | Overview of the Quantum Graph Transformer (QGT) architecture. Each input sentence $S$ is tokenized and represented as a graph $\mathcal{G}$ (either fully connected or $k$-nearest neighbor). Tokens are initialized directly with BERT embeddings ($d = 768$). These embeddings are processed through stacked `QTransformerConv` layers, followed by global mean pooling for graph-level representation. In each `QTransformerConv` layer, node features $\mathbf{x}_i$ are amplitude-encoded into quantum states $|\psi_i\rangle$, and processed by parameterized quantum circuits $U_Q(\boldsymbol{\theta})$ and $U_K(\boldsymbol{\phi})$ to generate query and key vectors. These quantum-encoded representations are used to compute attention scores $\alpha_{i,j}$, which guide message passing and node feature updates $\mathbf{x}'_i$. The pooled representation is passed through a classification head for label prediction. Model parameters *(highlighted in red)* are optimized end-to-end via backpropagation to minimize the cross-entropy loss.

stacked to progressively refine node representations and capture higher-order dependencies across the graph.

In today's AI-driven world, where improving natural language understanding is increasingly crucial, quantum-enhanced self-attention models have demonstrated significant potential in advancing NLP tasks. Inspired by this, we propose the *Quantum Graph Transformer (QGT)* model for sentiment classification, designed to capture deeper contextual dependencies through quantum-enhanced representations.

## 3 | Quantum Graph Transformer Model

Figure 1 illustrates the overall architecture of the proposed QGT model. The hybrid QGT architecture integrates quantum self-attention into a graph-based neural framework for sentiment classification. The model consists of the following key components.

### 3.1 | Sentence-to-Graph Construction

Each input sentence $S \in \mathcal{D}$, where $\mathcal{D}$ denotes the dataset and $S$ represents a single sentence, is first tokenized into a sequence of tokens. This sequence is denoted as $S = \{w_1, w_2, \ldots, w_N\}$, where each $w_i$ represents a token and $N$ denotes the total number of tokens in the sentence after tokenization. Based on this tokenized sequence, we construct a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ represents the tokens treated as nodes and $\mathcal{E}$ corresponds to the edge connections between tokens. In the QGT model, we employ two types of graph structures: (i) a *complete graph*, in which every node is connected to every other node ($(i, j) \in \mathcal{E}$ for all $i \neq j$); and (ii) a *k-nearest-neighbor (k-NN) graph*, where each node is connected only to its $k$ nearest tokens in the sequence. The complete graph enables dense connectivity, allowing each token to directly exchange information with all others, which is particularly effective for capturing long-range dependencies and global contextual cues. In contrast, the $k$-NN graph introduces sparsity,



25119044, 2026, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/qute.202501004 by Universita Di Salerno, Wiley Online Library on [02/07/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

focusing on local context while substantially reducing computational overhead, which is beneficial for shorter or more structured texts such as tweets, chat messages, and search queries. We note that the fully connected graph configuration is topologically equivalent to standard (non-graph) transformer attention, where every token attends to every other token. The key advantage of our graph-based formulation is the flexibility to employ sparse connectivity patterns such as $k$-NN, which reduces computational complexity from $\mathcal{O}(N^2)$ to $\mathcal{O}(Nk)$ while maintaining comparable performance (see Table 3). This framework thus generalizes both standard and sparse attention within a unified graph-based architecture.

### 3.2 | QTransformerConv: Embedding Encoding

Each node $i \in \mathcal{V}$ (corresponding to token $w_i$) is represented by an embedding vector $\mathbf{x}_i$. We evaluate QGT with two embedding types: (1) contextualized BERT embeddings ($\mathbf{x}_i \in \mathbb{R}^{768}$) extracted from the final hidden layer of a pretrained BERT-base model [30], which capture rich semantic and syntactic dependencies through deep bidirectional modeling; and (2) static GloVe embeddings ($\mathbf{x}_i \in \mathbb{R}^{50}$) [25], which represent words based on global word co-occurrence statistics. This evaluation allows us to compare the contribution of quantum attention learning with the embedding quality, as detailed in Section 4.4. To map these embeddings into the quantum domain, we employ *amplitude encoding*, an efficient data-loading scheme that embeds the components of a real-valued vector directly into the amplitude coefficients of a quantum state. Given a real-valued embedding vector for token $i$,

$$\mathbf{x}_i = [x_{i,0}, x_{i,1}, \ldots, x_{i,2^n-1}]^\top,$$

where each component $x_{i,j} \in \mathbb{R}$ represents the $j$-th element of the token embedding, amplitude encoding prepares an $n$-qubit quantum state as

$$|\psi_i\rangle = \frac{1}{\|\mathbf{x}_i\|_2} \sum_{j=0}^{2^n-1} x_{i,j} |j\rangle, \tag{5}$$

where each $x_{i,j}$ defines the amplitude of the computational basis state $|j\rangle$, and the normalization factor $\|\mathbf{x}_i\|_2$ ensures that $\langle\psi_i|\psi_i\rangle = 1$. In the QGT model, amplitude encoding is applied independently to each token's embedding. Since amplitude encoding requires the feature dimension to be a power of two, the embeddings are zero-padded to the nearest power of two: BERT embeddings (768-dimensional) are padded to $2^{10} = 1024$, requiring $n = 10$ qubits, while GloVe embeddings (50-dimensional) are padded to $2^6 = 64$, requiring $n = 6$ qubits. The resulting normalized vector is expressed as

$$\mathbf{x}_i = \frac{\tilde{\mathbf{x}}_i}{\|\tilde{\mathbf{x}}_i\|_2} = [x_{i,0}, x_{i,1}, \ldots, x_{i,2^n-1}]^\top, \quad \|\mathbf{x}_i\|_2 = 1.$$

We define the amplitude-encoding unitary $U_{\text{encoding}}(\mathbf{x}_i)$ that prepares the quantum state $|\psi_i\rangle$ from the initial all-zero state $|0\rangle^{\otimes n}$:

$$U_{\text{encoding}}(\mathbf{x}_i)|0\rangle^{\otimes n} = \sum_{j=0}^{2^n-1} x_{i,j}|j\rangle \equiv |\psi_i\rangle. \tag{6}$$

Here, $U_{\text{encoding}}(\mathbf{x}_i)$ represents the amplitude-loading transformation that maps the normalized classical embedding vector $\mathbf{x}_i$ onto the amplitudes of an $n$-qubit quantum state. In practice, this operation can be implemented using the `AmplitudeEmbedding` template provided by `PennyLane`, which efficiently initializes the quantum register such that the amplitudes correspond directly to the components of $\mathbf{x}_i$. Through this encoding, the semantic information contained in each token's embedding is compactly distributed across the amplitudes of the quantum state, thereby embedding the token representation into a high-dimensional *quantum feature space*. This representation provides an information-dense and hardware-efficient initialization for the subsequent parameterized quantum attention layer, as illustrated in Figure 2 *(left)*.

**Data-Loading Overhead and Scalability**. Amplitude encoding of high-dimensional embeddings requires $\mathcal{O}(2^n)$ operations for state preparation, which presents challenges for near-term quantum hardware. For BERT embeddings (10 qubits) and GloVe embeddings (6 qubits), this overhead is manageable in our classical simulations using PennyLane's `lightning.qubit` backend with adjoint differentiation. To further address hardware scalability, we employ a Quantum Autoencoder (QAE) [40] to compress high-dimensional quantum states into a smaller latent space. The QAE consists of an encoder circuit that maps the amplitude-encoded state to a compressed representation with fewer qubits, enabling deployment on more resource-constrained NISQ devices. As shown in Table 3, QAE-QGT (6 qubits) achieves competitive accuracy with only ∼3% degradation compared to the full 10-qubit BERT model, demonstrating a practical pathway for scaling QGT while remaining feasible on current quantum hardware.

### 3.3 | QTransformerConv: Quantum Self-Attention

After amplitude encoding, each node feature is represented as a quantum state $|\psi_i\rangle$ in a 10-qubit Hilbert space. We then apply a `QTransformerConv` layer, which performs message passing based on a quantum-enhanced self-attention mechanism. To compute query vectors for attention, we apply a parameterized quantum circuit $U_Q(\boldsymbol{\theta})$ consisting of $n$ qubits to each encoded state $|\psi_i\rangle$. As illustrated in Figure 2 *(right)*, the query circuit $U_Q(\boldsymbol{\theta})$ is composed of three layers of single-qubit rotations–$R_x$, $R_y$, and $R_z$–applied to every qubit, followed by a ring-style CNOT entanglement layer that introduces controlled correlations between adjacent qubits. This entanglement is succeeded by an additional layer of $R_y$ rotations and a Quantum Fourier Transform (QFT) [39] applied across all qubits to mix global phase information. A structurally identical circuit, $U_K(\boldsymbol{\phi})$, is used to generate key vectors, differing only in its parameter set $\boldsymbol{\phi}$. Both $U_Q(\boldsymbol{\theta})$ and $U_K(\boldsymbol{\phi})$ are fully differentiable quantum operators whose parameters are optimized jointly with the classical components of the network during training. Executing these circuits on $n$ qubits enables the model to capture higher-order dependencies among amplitude-encoded features within the quantum feature space.

The PQC architecture follows established designs from prior quantum self-attention work [14, 15]. The combination of hardware-efficient rotation layers with ring-style CNOT entan-



25119044, 2026, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/qute.202501004 by Universita Di Salerno, Wiley Online Library on [02/07/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

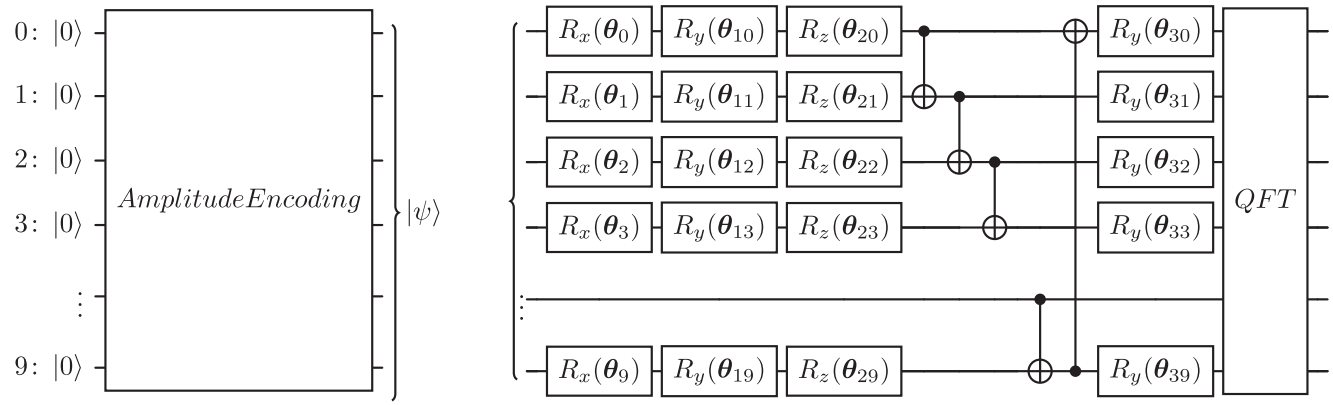



**FIGURE 2** | (*Left*) Quantum amplitude encoding circuit $U_{\text{encoding}}(\mathbf{x}_i)$ maps a BERT embedding vector $\mathbf{x}_i \in \mathbb{R}^{768}$ into a 10-qubit quantum state $|\psi_i\rangle$. The 768-dimensional embedding is padded to $2^{10} = 1024$ components, normalized to unit $\ell_2$-norm, and encoded into the amplitude coefficients of $|\psi_i\rangle$, providing a dense quantum representation of the token feature. (*Right*) Parameterized quantum circuit $U_Q(\boldsymbol{\theta})$ used for query vector generation consists of three layers of single-qubit rotations ($R_x$, $R_y$, and $R_z$) applied to each qubit, followed by a ring-style CNOT entanglement layer, an additional $R_y$ rotation layer, and a Quantum Fourier Transform (QFT) [39] across all qubits. The combined operation $U_Q(\boldsymbol{\theta}) \cdot U_{\text{encoding}}(\mathbf{x}_i)$ produces the query vector $\mathbf{Q}_i$ via Pauli X, Y and Z measurements, while a structurally identical circuit $U_K(\boldsymbol{\phi})$ generates the key vector $\mathbf{K}_i$ using a separate parameter set $\boldsymbol{\phi}$.

glement provides a balance between expressibility [41] and trainability [42] of the ansatz: (1) the rotation layers span the full single-qubit Bloch sphere, (2) ring entanglement avoids the barren plateau issues associated with highly entangled random circuits while still enabling non-trivial correlations, and (3) the QFT layer introduces global mixing without additional trainable parameters. Following Sim et al. [41], we compute expressibility as the KL divergence between the sampled fidelity distribution of $U_Q(\boldsymbol{\theta})$ and $U_K(\boldsymbol{\phi})$ and the Haar-random distribution, obtaining an expressibility of 0.107, confirming a highly expressible circuit for the QGT model.

The resulting query $\mathbf{Q}_i$ and key $\mathbf{K}_i$ representations are obtained by measuring the expectation values of Pauli operators on each qubit:

$$\mathbf{Q}_i = \left( \langle \psi_i | U_Q^\dagger(\boldsymbol{\theta})\, \sigma_q^{(p)}\, U_Q(\boldsymbol{\theta}) | \psi_i \rangle \right)_{\substack{q=0,\ldots,n-1 \\ p\in\{X,Y,Z\}}}, \tag{7}$$

$$\mathbf{K}_i = \left( \langle \psi_i | U_K^\dagger(\boldsymbol{\phi})\, \sigma_q^{(p)}\, U_K(\boldsymbol{\phi}) | \psi_i \rangle \right)_{\substack{q=0,\ldots,n-1 \\ p\in\{X,Y,Z\}}}. \tag{8}$$

In this work, we employ the Pauli $X$, $Y$, and $Z$ operators as the measurement observables on each qubit, producing three expectation values per qubit to form the query and key feature vectors of dimension $3n$. For a graph with $N$ nodes, this process is performed independently for each node to obtain its corresponding query and key quantum representations. The quantum attention coefficients are then computed analogously to the classical scaled dot-product attention used in Equations (2)–(3):

$$\alpha_{i,j} = \frac{\exp\left( \frac{\mathbf{Q}_i^\top \mathbf{K}_j}{\sqrt{q_d}} \right)}{\sum_{m\in\mathcal{N}(i)} \exp\left( \frac{\mathbf{Q}_i^\top \mathbf{K}_m}{\sqrt{q_d}} \right)}, \tag{9}$$

where $\alpha_{i,j}$ determines the influence of node $j$ on node $i$ during message passing and $q_d = 3n$ denotes the dimension of the query and key vectors (i.e., $q_d = 30$ for $n = 10$ qubits), corresponding to three Pauli expectation values per qubit. To enhance representational capacity, multi-headed quantum attention can be implemented by allocating separate parameter sets $\boldsymbol{\theta}^{(h)}$ and $\boldsymbol{\phi}^{(h)}$ for each head $h$, with each head operating on its own n-qubit parameterized circuits $U_Q^{(h)}(\boldsymbol{\theta}^{(h)})$ and $U_K^{(h)}(\boldsymbol{\phi}^{(h)})$.

### 3.4 | QTransformerConv: Message Passing

Once the quantum attention scores between token nodes are computed, the model performs message passing over the graph to propagate contextual information. Quantum-enhanced attention values from neighboring tokens are aggregated to update each node's embedding vector. Specifically, each token node sends messages to its directly connected neighbors (based on the edge set $\mathcal{E}$), with each message weighted by the quantum attention scores $\alpha_{i,j}$ defined in Equation 9. For each node $i$, the message received from a neighboring node $j \in \mathcal{N}(i)$ is computed as

$$m_{i,j} = \alpha_{i,j} \mathbf{x}_j,$$

where $\mathbf{x}_j$ denotes the classical embedding (value vector) of node $j$, which is directly used in message aggregation without an additional linear transformation. The total message aggregated at node $i$ from all of its neighbors is

$$m_i = \sum_{j\in\mathcal{N}(i)} m_{i,j}.$$

The node's feature vector is then updated by combining its original embedding $\mathbf{x}_i$ with the aggregated message:

$$\mathbf{x}_i' = \mathbf{x}_i + m_i.$$

This update scheme preserves the identity of each node's embedding while enriching it with context-aware features derived from its neighborhood using quantum attention weights.



25119044, 2026, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/qute.202501004 by Universita Di Salerno, Wiley Online Library on [02/07/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

### 3.5 | Classification Head

After the `QTransformerConv` layer refines the token (node) representations using quantum attention-guided message passing, the model aggregates node-level information for sentence-level classification. A global mean pooling operation is applied across all node embeddings $\mathbf{x}'_i$, producing a graph-level representation:

$$\mathbf{x}'_{\text{graph}} = \frac{1}{N}\sum_{i=1}^{N}\mathbf{x}'_i, \tag{10}$$

where $N$ is the number of nodes. This pooled representation is then passed through a fully connected layer to generate raw class logits $\mathbf{z}$, given by

$$\mathbf{z} = \mathbf{W}^{\top}\mathbf{x}'_{\text{graph}} + \mathbf{b}, \tag{11}$$

where $\mathbf{W}$ is the weight matrix and $\mathbf{b}$ is the bias vector. The logits $\mathbf{z}$ are used in the cross-entropy loss [43], which internally applies the softmax function to compute the probability distribution over classes. The loss is defined as

$$\mathcal{L} = -\sum_{i=1}^{c} y_i \log\left(\frac{\exp(z_i)}{\sum_{j=1}^{c}\exp(z_j)}\right), \tag{12}$$

where $c$ is the number of classes, $y_i$ is the true label, and $z_i$ is the logit for class $i$. During backpropagation, the model's trainable parameters are updated to minimize this loss. Figure 1 shows the trainable parameters in red. The learnable parameters originate from two main components: (1) the quantum self-attention mechanism, which includes $U_Q(\boldsymbol{\theta})$ and $U_K(\boldsymbol{\phi})$, each containing $n \times 4$ trainable parameters corresponding to ten qubits and four rotation gates per layer; and (2) the classical classification head, implemented as a fully connected layer.

### 3.6 | Loss Function and Regularization

To encourage stronger gradient updates to the PQC parameters during training, we incorporate a scaling term and attention regularization. The total loss function is defined as:

$$\mathcal{L}_{\text{total}} = (1 + \lambda_{\text{amp}}) \cdot \mathcal{L}_{\text{CE}} + \lambda_{\text{ent}} \cdot \mathcal{L}_{\text{ent}}, \tag{13}$$

where $\mathcal{L}_{\text{CE}}$ is the cross-entropy loss and $\lambda_{\text{amp}} = 0.1$ amplifies the loss signal, encouraging stronger gradient updates to the PQC parameters during training. Additionally, we include an attention entropy regularizer:

$$\mathcal{L}_{\text{ent}} = -\sum_{i}\sum_{j\in\mathcal{N}(i)}\alpha_{i,j}\log\alpha_{i,j}, \tag{14}$$

with $\lambda_{\text{ent}} = 0.01$, which encourages diverse attention distributions and prevents collapse to uniform or degenerate patterns. As shown in Table 2 a, disabling the loss amplification reduces accuracy by 2.1%, confirming its contribution to model performance.

## 4 | Experimental Result and Analysis

### 4.1 | Experimental Setup

We evaluate the QGT model on five commonly used benchmark datasets for QNLP models. We selected these datasets because all previous QNLP works have been evaluated on them, enabling direct comparison with existing methods. Three datasets are from YELP, IMDB, and AMAZON, consisting of 1,000 samples each [21]. The YELP dataset contains restaurant reviews labeled with positive or negative sentiment, the IMDB dataset consists of movie reviews labeled with positive or negative sentiment, and the AMAZON dataset includes user product reviews with binary ratings of 0 or 1. The three datasets have maximum sentence lengths of 34, 45, and 30, respectively. The other two datasets, MC (Meaning Classification) and RP (Relative Pronoun), are synthetic and generated as described in [11]. They have relatively smaller vocabularies, and each sample has a length of 4 tokens. The MC dataset contains 130 samples, while the RP dataset has 105 samples. Each of the five datasets is split into 70%, 10%, and 20% as training, validation, and testing sets, respectively, and a separate model is trained for each dataset. Training is performed using the PennyLane Lightning device [44], and the model is run on macOS with an Apple M3 Max chip and 64 GB of RAM.

Each dataset is processed and trained using the QGT model described in Section 3. In our experiments, we use $n = 10$ qubits for BERT embedding and $n = 6$ qubits for GloVe embedding with a single layer of PQCs, $U_Q(\boldsymbol{\theta})$ and $U_K(\boldsymbol{\phi})$, to compute the query and key vectors, respectively. The parameters $\boldsymbol{\theta}$ and $\boldsymbol{\phi}$ are initialized from a normal distribution with mean 0 and standard deviation 0.01. We employ the Adam optimizer with a learning

**TABLE 1** | Training hyperparameters.

| Hyperparameter | Value | Hyperparameter | Value |
|---|---|---|---|
| *Model Architecture* | | *Quantum Circuit* | |
| Embedding dim | 768 | Quantum LR multiplier | 5.0× |
| Number of qubits | 10 | Q/K temperature mode | Learned |
| Amplitude dim | 1024 | Q/K temperature init | 5.0 |
| GNN layers | 1 | Residual mix init | 1.0 |
| Q/K dimension | 30 | Entanglement type | Ring CNOT |
| *Training* | | *Regularization* | |
| Batch size | 32 | Loss amplification | Enabled |
| Epochs | 25 | $\lambda_{\text{amp}}$ | 0.1 |
| Learning rate | 0.01 | Attention entropy $\lambda$ | 0.01 |
| Gradient clipping | 1.0 | Freeze FC epochs | 3 |
| Scheduler step | 5 | | |
| Scheduler gamma | 0.7 | | |
| Early stopping | 3 | | |



25119044, 2026, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/qute.202501004 by Universita Di Salerno, Wiley Online Library on [02/07/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

**TABLE 2** | Ablation study results and parameter count comparison.

| (a) Ablation Study (Yelp, BERT, Fully connected) | | (b) Parameter Count (Using BERT Embedding) | | | |
|---|---|---|---|---|---|
| **Configuration** | **Test Acc. (%)** | **Model** | **Total** | **Quantum** | **Classical Q/K** |
| Full QGT (10-qubit) | 92.25 ± 0.29 | QGT (10-qubit) | 1,620 | 80 | 0 |
| No QFT | 90.16 ± 0.29 | QAE-QGT (6-qubit) | 5,488 | 108 | 0 |
| Linear Entanglement | 89.83 ± 0.29 | CGT | 47,620 | 0 | 46,080 |
| No Loss Amplification | 90.16 ± 0.29 | 29.4× fewer params: QGT vs CGT | | | |
| No QFT & No Amp. | 89.16 ± 0.29 | 8.7× fewer params: QAE-QGT vs CGT | | | |
| QAE-QGT (6-qubit) | 89.25 ± 0.29 | | | | |

**TABLE 3** | Performance comparison (accuracy %) of QGT and other attention-based models across five datasets. Values are reported as mean ± standard deviation across four independent runs.

| Model | YELP | AMAZON | IMDB | MC | RP |
|---|---|---|---|---|---|
| *Our Models (BERT Embedding)* | | | | | |
| QGT (Full) | 92.25 ± 0.29 | 91.25 ± 0.29 | 91.38 ± 0.25 | 100.00 ± 0.00 | 100.00 ± 0.00 |
| QGT ($k$-NN: 4) | 87.38 ± 0.25 | 88.12 ± 0.48 | 88.38 ± 0.25 | 100.00 ± 0.00 | 95.38 ± 0.25 |
| QGT ($k$-NN: 5) | 91.38 ± 0.25 | 91.25 ± 0.29 | 90.38 ± 0.25 | 100.00 ± 0.00 | 100.00 ± 0.00 |
| QAE-QGT (Full, 6 qubit) | 89.25 ± 0.29 | 88.25 ± 0.29 | 89.38 ± 0.25 | 100.00 ± 0.00 | 95.00 ± 0.00 |
| CGT (Full) | 89.25 ± 0.29 | 89.25 ± 0.29 | 88.62 ± 0.48 | 100.00 ± 0.00 | 81.82 ± 0.00 |
| *Our Models (GloVe Embedding)* | | | | | |
| QGT (Full, GloVe) | 82.50 ± 0.00 | 83.50 ± 0.00 | 75.00 ± 0.00 | 100.00 ± 0.00 | 63.64 ± 0.00 |
| CGT (Full, GloVe) | 77.00 ± 0.00 | 84.50 ± 0.00 | 73.00 ± 0.00 | 100.00 ± 0.00 | 63.64 ± 0.00 |
| *Prior Quantum Attention Models* | | | | | |
| QSAM (Zhang et al., 2024) [16] | 91.03 ± 0.94 | 85.73 ± 1.81 | 88.35 ± 1.16 | — | — |
| QMSAN (Chen et al., 2025) [15] | 84.96 ± 3.34 | 87.48 ± 1.02 | 84.82 ± 2.96 | 100.00 | 75.63 ± 1.60 |
| QSANN (Li et al., 2024) [14] | 84.79 ± 1.29 | 84.25 ± 1.75 | 80.28 ± 1.78 | 100.00 | 67.74 |

rate of 0.01 and use the cross-entropy loss [43], which is well-suited for classification tasks involving categorical sentiment labels. A StepLR scheduler is applied with a step size of 5 and decay factor ($\gamma = 0.7$), along with a batch size of 32. Early stopping is used based on validation loss with a patience of 3 epochs, with training running for up to 25 epochs. Additionally, we incorporate a loss amplification term with $\lambda_{\text{amp}} = 0.1$ to encourage stronger gradient updates to the PQC parameters. We train the model using both fully connected and $k$-nearest-neighbor ($k$-NN) graph structures, allowing us to evaluate the QGT's ability to capture both global and local contextual dependencies. We also experimented with multiple PQC layers and multi-headed quantum attention; however, no significant improvement was observed on these datasets, likely due to their relatively small size. A complete list of hyperparameters is provided in Table 1.

To ensure a fair comparison between quantum and classical approaches, we implement a CGT baseline with an architecture identical to QGT, differing only in the attention computation mechanism. Both models share the same graph construction method, input embeddings, number of transformer convolution layers, classification head architecture, output dimensions, and training configuration (see Table 1). The sole difference lies in the attention mechanism: CGT uses classical dot-product attention with learned projection matrices $\mathbf{W}_Q, \mathbf{W}_K \in \mathbb{R}^{768\times768}$, while QGT replaces these with parameterized quantum circuits $U_Q(\boldsymbol{\theta})$ and $U_K(\boldsymbol{\phi})$. This controlled setup isolates the attention mechanism as the only variable, enabling direct evaluation of the quantum component's contribution to model performance.

## 4.2 | Model Learning Dynamics

We first analyze the learning dynamics of the proposed QGT model to evaluate its performance. Figure 3 *(left)* presents the training loss curves for all five datasets using BERT embeddings. The plots show a consistent reduction in training loss over epochs, indicating effective learning across all datasets. In each case, the loss decreases sharply during the initial epochs, followed by gradual convergence. Figure 3 *(right)* shows the validation



25119044, 2026, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/qute.202501004 by Universita Di Salerno, Wiley Online Library on [02/07/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

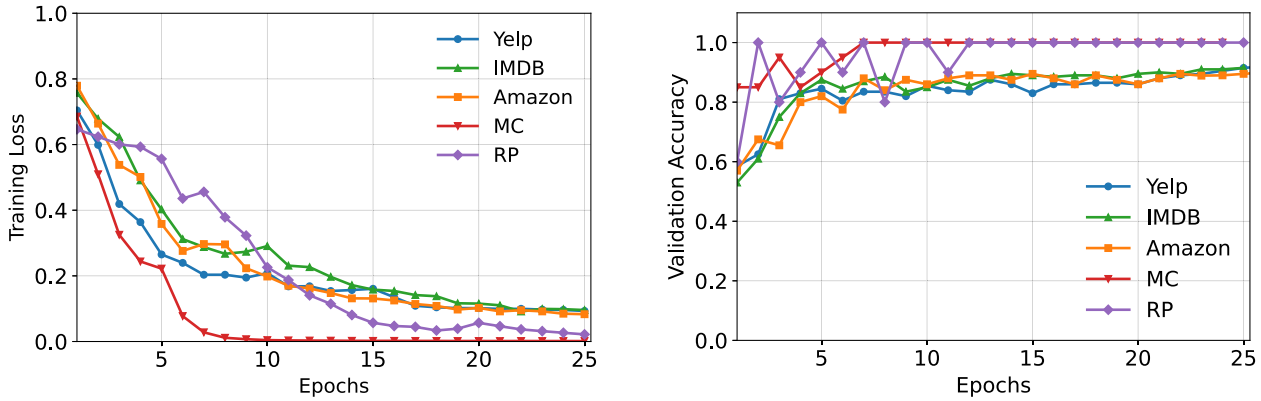



**FIGURE 3** | *(Left)* Training loss curves for the five datasets in the sentiment classification task using the proposed QGT model with BERT embedding, demonstrating consistent learning across datasets. *(Right)* Validation accuracy for the five datasets, showing stable convergence and strong generalization performance across both real-world and synthetic benchmarks.

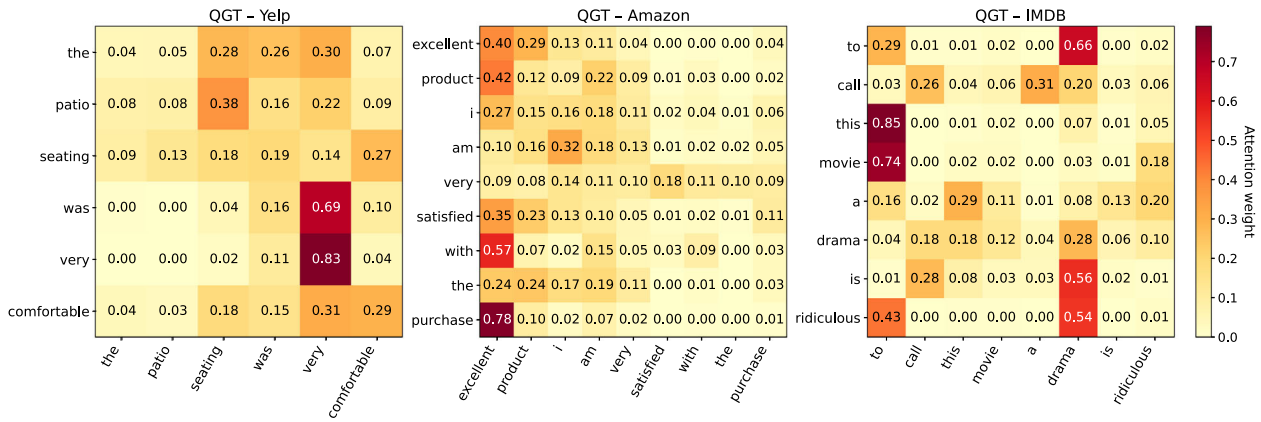



**FIGURE 4** | Attention heatmaps for the QGT model using BERT embedding across the YELP, AMAZON, and IMDB sentiment datasets. Each cell visualizes token-to-token attention weights, where brighter colors indicate stronger focus. The plot shows QGT learned attentions exhibit localized patterns concentrated around sentiment-bearing tokens.

accuracy curves for the same datasets. The validation accuracy closely follows the training accuracy, demonstrating the model's ability to generalize while minimizing loss over epochs. We further examine the attention distributions learned by QGT. Figure 4 shows the attention maps produced by QGT on the YELP, AMAZON, and IMDB datasets, which exhibit sharp, sentiment-focused peaks. For example, in YELP (*"the patio seating was very comfortable"*), multiple tokens attend strongly to the intensifier *very* (e.g., `was` → `very`: 0.69; `comfortable` → `very`: 0.31). Similarly, in AMAZON (*"excellent product i am very satisfied with the purchase"*), the attention mass concentrates on the sentiment-bearing token *excellent* (e.g., `product` → `excellent`: 0.42; `purchase` → `excellent`: 0.78). These results demonstrate that QGT selectively focuses on sentiment-relevant tokens, learning interpretable attention patterns that align with human intuition about sentiment-bearing words.

### 4.3 | Hyperparameter Sensitivity and Ablation Study

Table 1 lists all hyperparameters used in our experiments. To evaluate the contribution of each quantum component, we conduct ablation studies on the YELP dataset shown in Table 2 a. For all ablation experiments, hyperparameters including learning rate were kept constant to isolate the effect of each component. Removing the QFT layer reduces accuracy by 2.1%, replacing ring with linear entanglement reduces accuracy by 2.4%, and disabling loss amplification reduces accuracy by 2.1%. We note that the accuracy reduction when disabling loss amplification may partially reflect suboptimal learning dynamics at the fixed learning rate; however, the consistent improvement when amplification is enabled suggests it provides a beneficial inductive bias for PQC training. These results confirm that each component contributes meaningfully to QGT's performance.

Table 2 b compares parameter counts across models using BERT embedding. QGT (10-qubit) requires only 1620 parameters (80 quantum), compared to CGT's 47 620 parameters (46 080 for classical Q/K projections), representing a **29.4**× reduction while achieving superior accuracy. To address scalability for higher-dimensional embeddings on near-term quantum devices, we employ a Quantum Autoencoder (QAE) to compress 10-qubit circuits to 6 qubits. QAE-QGT achieves **8.7**× fewer parameters than CGT with only modest accuracy loss (~3%), demonstrating a practical pathway for scaling QGT to larger embedding dimensions while remaining feasible on current quantum hardware.



25119044, 2026, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/qute.202501004 by Universita Di Salerno, Wiley Online Library on [02/07/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

**TABLE 4** | Performance comparison (accuracy %) of QGT and prior non-attention quantum NLP models across five datasets. Values for QGT are reported as mean ± standard deviation across four independent runs.

| Model (Others) | YELP | AMAZON | IMDB | MC | RP |
|---|---|---|---|---|---|
| QGT (Full) | 92.25 ± 0.29 | 91.25 ± 0.29 | 91.38 ± 0.25 | 100.00 ± 0.00 | 100.00 ± 0.00 |
| LEXIQL (Silver et al., 2024) | 89.0 | 81.0 | 92.0 | 100.0 | 64.0 |
| DisCoCat (Kartsaklis et al., 2021) | 73.0 | 61.0 | 62.0 | 100.0 | 68.0 |
| AMPMEAN (Alexander et al., 2022) | 44.0 | 41.0 | 55.0 | 87.0 | 61.0 |

### 4.4 | Comparison with Existing Quantum NLP Models

Several quantum self-attention mechanisms have been proposed for NLP tasks. QSAM [16] employs quantum kernel methods to compute attention similarity but operates on fixed sequence structures without graph connectivity. QMSAN [15] uses mixed-state representations for attention computation, incurring higher qubit overhead. QSANN [14] implements quantum self-attention but is limited to sequential token structures. In contrast, QGT introduces several novel features: (1) integration of quantum attention with graph neural network message passing, allowing flexible topology control; (2) support for both fully connected ($\mathcal{O}(N^2)$) and sparse $k$-NN ($\mathcal{O}(Nk)$) connectivity; (3) superior parameter efficiency with only 80 trainable quantum parameters; and (4) leveraging highly contextual embeddings such as BERT as a strong initialization, providing a richer semantic foundation for the quantum attention mechanism. These design choices enable QGT to achieve state-of-the-art performance while maintaining scalability for longer sequences.

We compare the performance of the proposed QGT model with existing QNLP models developed for sentiment classification tasks. Table 3 summarizes the test accuracy of the QGT model evaluated with both fully connected and $k$-NN graph configurations, together with other attention-based quantum models, including QSAM [16], QMSAN [15], and QSANN [14], across five benchmark datasets. Results from a Classical Graph Transformer (CGT) model with identical architecture and training configuration (see Section 4.1) are also included to isolate the contribution of the quantum attention mechanism. The QGT model with BERT embedding consistently outperforms all other attention-based models across the five datasets, achieving the highest average accuracy in every case. Notably, the fully connected QGT achieves accuracies of 92.25%, 91.25%, and 91.38% on YELP, AMAZON, and IMDB, respectively, while maintaining perfect accuracy on the synthetic MC and RP datasets. Compared to CGT, QGT achieves an average improvement of 2.6% on real-world datasets and 18.2% on the RP dataset, demonstrating the benefit of quantum attention. Even when using sparse $k$-NN connectivity (for example, $k = 5$), QGT maintains comparable performance, indicating its robustness and efficiency in learning semantic relationships under constrained graph structures. Although the fully connected graph configuration requires $\mathcal{O}(N^2)$ edges for $N$ tokens, comparable performance can be achieved even with sparse $k$-NN connectivity, which scales as $\mathcal{O}(Nk)$. For example, the $k$-NN ($k = 5$) variant attains nearly identical accuracy to the full model across all datasets. We find that for this benchmark set, QGT efficiently captures long-range semantic dependencies even with reduced computational complexity; however, longer or more complex sentences may require higher connectivity to preserve global context.

Table 4 further compares QGT with prior non-attention QNLP baselines, including the grammar-aware DISCOCAT model [11, 24], the ensemble-based LEXIQL [17], and the amplitude-based AMPMEAN model [12]. QGT surpasses all these baselines on four of the five datasets, with only a slight performance difference from LEXIQL on IMDB. Overall, QGT demonstrates superior generalization across both real-world and synthetic benchmarks, outperforming classical and quantum baselines alike.

To further isolate the contribution of the quantum attention mechanism from the pre-trained embeddings, we conduct experiments using GloVe embeddings (50-dimensional) instead of BERT. As shown in Table 3, QGT (GloVe) outperforms or matches CGT (GloVe) on four of five datasets, with particularly strong improvement on YELP (+5.5%) and IMDB (+2.0%). This confirms that the quantum advantage persists even without highly contextual embeddings, demonstrating that the performance gains are attributable to the quantum attention mechanism itself. The results also highlight that using contextual embeddings like BERT provides a stronger starting point for achieving optimal performance.

### 4.5 | Quantum NLP Advantage

The QGT model leverages parameterized quantum circuits (PQCs) to compute attention with significantly fewer parameters compared to the classical attention mechanism. While classical attention layers require high-dimensional projections with large weight matrices, our QGT model generates query and key vectors using only $4n$ trainable parameters for each of $U_Q(\boldsymbol{\theta})$ and $U_K(\boldsymbol{\phi})$, resulting in a total of 80 parameters with a 10-qubit circuit for BERT embedding. Despite this drastic parameter reduction, the comparison in Table 3 shows that the QGT model effectively learns meaningful attention patterns and outperforms the classical graph transformer model. The key to this efficiency lies in the expressive power of PQCs, which enables the model to capture rich quantum representations within the high-dimensional Hilbert space.

One of the major limitations of current classical NLP models is their reliance on large corpora of data and computationally intensive architectures. Any viable alternative model should aim to overcome these constraints. To evaluate sample efficiency, we vary the amount of training data from 10% to 100% while keeping



25119044, 2026, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/qute.202501004 by Universita Di Salerno, Wiley Online Library on [02/07/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

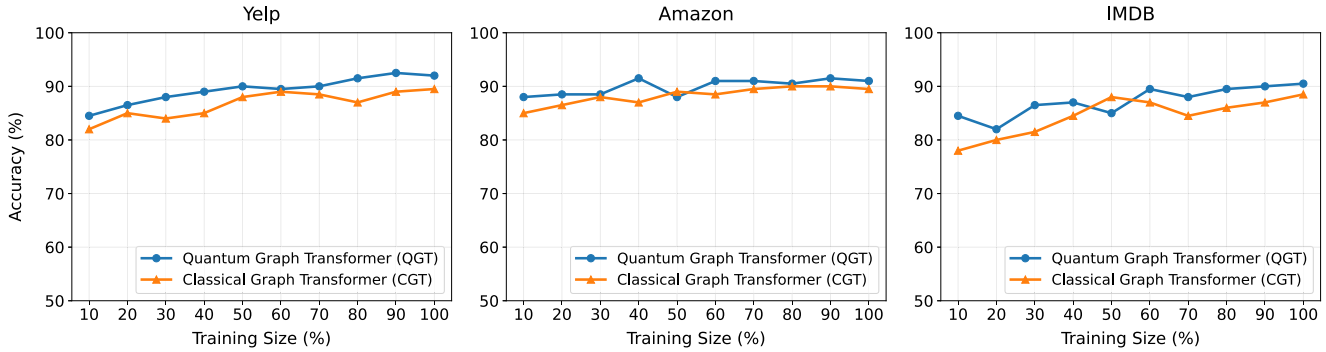



**FIGURE 5** | Comparison of sample efficiency across three datasets. Each curve shows test accuracy versus training sample size for QGT and CGT. QGT consistently outperforms CGT across all training fractions on YELP, AMAZON, and IMDB datasets, demonstrating better sample efficiency.

the test set fixed. Figure 5 presents the testing accuracy performance of both the quantum and classical graph transformer models across varying training sample sizes for the YELP *(left)*, AMAZON *(middle)*, and IMDB *(right)* datasets. The plots show that QGT consistently outperforms CGT across most training fractions. At 10% training data, QGT achieves 84.5%, 88.0%, and 84.5% accuracy on YELP, AMAZON, and IMDB respectively, compared to CGT's 82.0%, 85.0%, and 78.0%. This advantage persists at full training data, demonstrating that QGT requires 3–5× fewer samples to reach comparable performance. These results collectively demonstrate that across all three datasets, the QGT architecture exhibits sample efficiency, achieving comparable or better performance with substantially fewer training samples.

## 5 | Conclusion

In this work, we presented the Quantum Graph Transformer (QGT), a hybrid quantum–classical architecture that integrates quantum self-attention into a graph-based neural framework for sentiment classification. The model performs message passing through attention scores computed via parameterized quantum circuits, effectively capturing complex contextual dependencies between tokens. Experimental results across five benchmark datasets demonstrate that QGT achieves state-of-the-art performance among existing QNLP models and outperforms a classical graph transformer baseline with identical architecture.

Despite these promising results, the current evaluation is limited to relatively small-scale and synthetic datasets. Due to the small dataset sizes, the accuracy improvement of QGT over CGT is modest (∼2.6% on real-world datasets). However, evaluating QGT on larger benchmarks may reveal greater performance gains where the expressive power of quantum attention can be more fully utilized. Future extensions of this work should investigate the scalability and generalization of QGT on larger and more diverse benchmarks such as SST-5 [45], TWITTER SENTIMENT [46], and the Kaggle IMDB Movie Review dataset [47]. To adapt to such datasets, deeper quantum architectures incorporating multiple layers of query and key PQCs, as well as multi-head quantum self-attention mechanisms, could be explored to capture richer semantic structures. Since each attention head requires a separate quantum circuit, this would increase the overall qubit requirements. Additionally, future research should examine the robustness of QGT under realistic noise conditions and explore optimized graph topologies that leverage semantic or grammar-aware token connections. Overall, these directions can further advance the development of scalable, noise-resilient, and data-efficient quantum models for natural language understanding.


### Acknowledgements

The research presented in this article was supported by the NNSA's Advanced Simulation and Computing Beyond Moore's Law program at Los Alamos National Laboratory and the Laboratory Directed Research and Development program of Los Alamos National Laboratory under project number 20230049DR. This work has been assigned LANL technical report number LA-UR-25-31321.


### Data Availability Statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

### Conflicts of Interest

The authors declare no conflict of interest.

### References


1. S. J. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th edn. (Pearson, 2016).

2. S. B. B. Priyadarshini, A. B. Bagjadab, and B. K. Mishra, *Natural Language Processing in Artificial Intelligence* (Apple Academic Press, 2020), ch. A Brief Overview of Natural Language Processing and Artificial Intelligence, 211–224.

3. Y. Moslem, R. Haque, J. D. Kelleher, and A. Way, "Adaptive Machine Translation with Large Language Models," in *24th Annual Conference of the European Association for Machine Translation* (European Association for Machine Translation, 2023),227–237.

4. K. L. Tan, C. P. Lee, and K. M. Lim, "A Survey of Sentiment Analysis: Approaches, Datasets, and Future Research," *Applied Sciences* 13, no. 7 (2023): 4550.

5. OpenAI, J. Achiam, S. Adler, S. Agarwal, et al., "GPT-4 Technical Report," *arXiv preprint* 1–100 (2024).

6. X. Ding, L. Chen, M. Emani, et al., "HPC-GPT: Integrating Large Language Model for High-Performance Computing," in *SC-W'23 Workshops of The International Conference on High Performance Computing, Network, Storage, and Analysis* (Association for Computing Machinery, 2023),951–960.





25119044, 2026, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/qute.202501004 by Universita Di Salerno, Wiley Online Library on [02/07/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

7. J. Biamonte, P. Wittek, N. Pancotti, P. Rebentrost, N. Wiebe, and S. Lloyd, "Quantum Machine Learning," *Nature* 549 (2017): 195–202.

8. M. Cerezo, G. Verdon, H.-Y. Huang, L. Cincio, and P. J. Coles, "Challenges and Opportunities in Quantum Machine Learning," *Nature Computational Science* 2 (2022): 567–576.

9. M. Schuld and N. Killoran, "Quantum Machine Learning in Feature Hilbert Spaces," *Physical Review Letters* 122, no. 4 (2019): 040504.

10. V. Havlíček, A. D. Córcoles, K. Temme, et al., "Supervised Learning with Quantum-Enhanced Feature Spaces," *Nature* 567 (2019): 209–212.

11. R. Lorenz, A. Pearson, K. Meichanetzidis, D. Kartsaklis, and B. Coecke, "QNLP in Practice: Running Compositional Models of Meaning on a Quantum Computer," *Journal of Artificial Intelligence Research* 76 (2023): 1305–1342.

12. A. Alexander and D. Widdows, "Quantum Text Encoding for Classification Tasks," in *2022 IEEE/ACM 7th Symposium on Edge Computing (SEC)* (IEEE, 2022),355–361.

13. J. Stein, I. Christ, N. Kraus, M. B. Mansky, R. Müller, and C. Linnhoff-Popien, "Applying QNLP to Sentiment Analysis in Finance," in *2023 IEEE International Conference on Quantum Computing and Engineering (QCE)* (IEEE, 2023), 20–25.

14. G. Li, X. Zhao, and X. Wang, "Quantum Self-Attention Neural Networks for Text Classification," *Science China Information Sciences* 67 (2024): 142501.

15. F. Chen, Q. Zhao, L. Feng, C. Chen, Y. Lin, and J. Lin, "Quantum Mixed-State Self-Attention Network," *Neural Networks* 185 (2025): 107123.

16. H. Zhang, Q. Zhao, and C. Chen, "A Light-Weight Quantum Self-Attention Model for Classical Data Classification," *Applied Intelligence* 54, no. 4 (2024): 3077–3091.

17. D. Silver, A. Ranjan, R. Achutha, T. Patel, and D. Tiwari, "LEX-IQL: Quantum Natural Language Processing on NISQ-Era Machines," in *SC'24: International Conference for High Performance Computing, Networking, Storage and Analysis*, IEEE (IEEE, 2024), 1–15.

18. Z. Wu, S. Pan, F. Chen, G. Long, C. Zhang, and P. S. Yu, "A Comprehensive Survey on Graph Neural Networks," *IEEE Transactions on Neural Networks and Learning Systems* 32, no. 1 (2021): 4–24.

19. B. Liu and L. Wu, *Graph Neural Networks: Foundations, Frontiers, and Applications* (Springer, 2022), ch. Graph Neural Networks in Natural Language Processing 463–481.

20. F. Scarselli, M. Gori, A. C. Tsoi, M. Hagenbuchner, and G. Monfardini, "The Graph Neural Network Model," *IEEE Transactions on Neural Networks* 20, no. 1 (2009): 61–80.

21. D. Kotzias, M. Denil, N. De Freitas, and P. Smyth, "From Group to Individual Labels Using Deep Features," in *21th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (Association for Computing Machinery, 2015),597–606.

22. Z. S. Harris, "Distributional Structure," *Word* 10, no. 2–3 (1954): 146–162.

23. P. F. Brown, V. J. Della Pietra, P. V. Desouza, J. C. Lai, and R. L. Mercer, "Class-Based N-Gram Models of Natural Language," *Computational Linguistics* 18, no. 4 (1992): 467–480.

24. K. W. Church, "Word2Vec," *Natural Language Engineering* 23, no. 1 (2017): 155–162.

25. J. Pennington, R. Socher, and C. Manning, "GloVe: Global Vectors for Word Representation," in *2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)* (Association for Computational Linguistics, 2014),1532–1543.

26. D. Tang, B. Qin, and T. Liu, "Document Modeling with Gated Recurrent Neural Network for Sentiment Classification," in *2015 Conference on Empirical Methods in Natural Language Processing (EMNLP)* (Association for Computational Linguistics, 2015),1422–1432.

27. S. Hochreiter and J. Schmidhuber, "Long Short-Term Memory," *Neural Computation* 9, no. 8 (1997): 1735–1780.

28. K. Cho, B. van Merriënboer, C. Gulcehre, et al., "Learning Phrase Representations Using RNN Encoder-Decoder for Statistical Machine Translation," in *2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)* (Association for Computational Linguistics, 2014),1724–1734.

29. A. Vaswani, N. Shazeer, N. Parmar, et al., "Attention Is All You Need," in *Advances in Neural Information Processing Systems 30: Annual Conference on Neural Information Processing Systems NIPS 2017* (Curran Associates, Inc., 2017),5998–6008.

30. J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-Training of Deep Bidirectional Transformers for Language Understanding," in *2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)* (Association for Computational Linguistics, 2019),4171–4186.

31. L. Floridi and M. Chiriatti, "GPT-3: Its Nature, Scope, Limits, and Consequences," *Minds and Machines* 30 (2020): 681–694.

32. Y. Liu, M. Ott, N. Goyal, et al., "RoBERTa: A Robustly Optimized BERT Pretraining Approach," *arXiv Preprint* (2019): 1–13.

33. C. M. Varmantchaonala, J. L. K. E. Fendji, J. Schöning, and M. Atemkeng, "Quantum Natural Language Processing: A Comprehensive Survey," *IEEE Access* (2024).

34. B. Coecke, M. Sadrzadeh, and S. Clark, "Mathematical Foundations for a Compositional Distributional Model of Meaning," *Linguistic Analysis* 36, no. 1–4 (2010): 345–384, festschrift for Joachim Lambek.

35. S. Ganguly, S. N. Morapakula, and L. M. P. Coronado, "Quantum Natural Language Processing Based Sentiment Analysis Using Lambeq Toolkit," in *2022 Second International Conference on Power, Control and Computing Technologies (ICPC2T)* (IEEE, 2022),1–6.

36. J. D. Castillo, D. Zhao, and Z. Pei, "Comparative Study of the Ansätze in Quantum Language Models," *Advanced Quantum Technologies* 8, no. 10 (2025): e2500134.

37. S. Aktar, A. Bärtschi, A.-H. A. Badawy, and S. Eidenben, "Quantum Graph Transformer for NLP Sentiment Classification," in *2025 IEEE International Conference on Quantum Computing and Engineering (QCE)* 02 (2025): 552–553.

38. S. Yun, M. Jeong, R. Kim, J. Kang, and H. J. Kim, "Graph Transformer Networks," in *Advances in Neural Information Processing Systems 32: Annual Conference on Neural Information Processing Systems (NeurIPS 2019)* (Curran Associates, Inc., 2019), 11 960–11 970.

39. Y. S. Weinstein, M. A. Pravia, E. M. Fortunato, S. Lloyd, and D. G. Cory, "Implementation of the Quantum Fourier Transform," *Physical Review Letters* 86, no. 9 (2001): 1889.

40. J. Romero, J. P. Olson, and A. Aspuru-Guzik, "Quantum Autoencoders for Efficient Compression of Quantum Data," *Quantum Science and Technology* 2, no. 4 (2017): 045001.

41. S. Sim, P. D. Johnson, and A. Aspuru-Guzik, "Expressibility and Entangling Capability of Parameterized Quantum Circuits for Hybrid Quantum-Classical Algorithms," *Advanced Quantum Technologies* 2, no. 12 (2019): 1900070.

42. J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, and H. Neven, "Barren Plateaus in Quantum Neural Network Training Landscapes," *Nature Communications* 9, no. 1 (2018): 4812.

43. A. Paszke, S. Gross, F. Massa, et al., "PyTorch: An Imperative Style, High-Performance Deep Learning Library," *Advances in Neural Information Processing Systems* 32 (2019).

44. V. Bergholm, J. Izaac, M. Schuld, et al., "PennyLane: Automatic Differentiation of Hybrid Quantum-Classical Computations," arXiv preprint (July 2022).

45. R. Socher, A. Perelygin, J. Wu, et al., "Recursive Deep Models for Semantic Compositionality Over a Sentiment Treebank," in *2013 Conference on Empirical Methods in Natural Language Process-*



25119044, 2026, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/qute.202501004 by Universita Di Salerno, Wiley Online Library on [02/07/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

*ing (EMNLP)* (Association for Computational Linguistics, 2013),1631–1642.

46. S. Rosenthal, N. Farra, and P. Nakov, "SemEval-2017 Task 4: Sentiment Analysis in Twitter," in *11th International Workshop on Semantic Evaluation (SemEval-2017)* (Association for Computational Linguistics, 2017),502–518.

47. L. Narayan, "IMDB Dataset of 50K Movie Reviews," 2018, accessed: 2025-05-14.



25119044, 2026, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/qute.202501004 by Universita Di Salerno, Wiley Online Library on [02/07/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License